is given by $= \frac{\text{Energy}}{\text{speed}} = \frac{U}{c}$. The force exerted by an electromagnetic wave on unit area of a surface is called radiation pressure.

12. If the electromagnetic wave incident on a material surface is completely absorbed, then the energy delivered is U and momentum imparted on the surface is $p = \frac{U}{c}$.
13. If the incident electromagnetic wave of energy U is totally reflected from the surface, then the momentum delivered to the surface is $\Delta p = \frac{U}{c} - \left(-\frac{U}{c}\right) = 2\frac{U}{c}$.
14. The rate of flow of energy crossing a unit area is known as pointing vector for electromagnetic waves, which is $\vec{S} = \frac{1}{\mu_\circ}\left(\vec{E} \times \vec{B}\right) = c^2 \varepsilon_\circ \left(\vec{E} \times \vec{B}\right)$. The unit for pointing vector is W m^{-2}. The pointing vector at any point gives the direction of energy transport from that point.
15. Electromagnetic waves carries not only energy and momentum but also angular momentum.

EXAMPLE 5.2

The relative magnetic permeability of the medium is 2.5 and the relative electrical permittivity of the medium is 2.25. Compute the refractive index of the medium.

Solution

Dielectric constant (relative permeability of the medium) is $\varepsilon_r = 2.25$

Magnetic permeability is $\mu_r = 2.5$

Refractive index of the medium,

$$n = \sqrt{\varepsilon_r \mu_r} = \sqrt{2.25 \times 2.5} = 2.37$$

5.2.2 Sources of electromagnetic waves

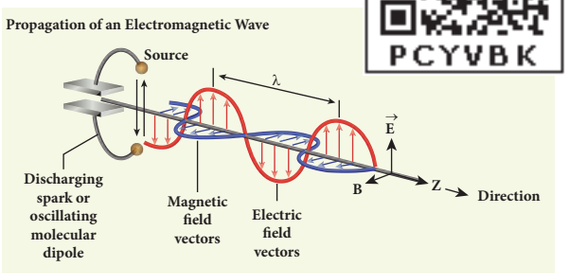


Figure 5.9 Oscillating charges - sources of electromagnetic waves

Any stationary source charge produces only electric field (refer Unit 1). When the charge moves with uniform velocity, it produces steady current which gives rise to magnetic field (not time dependent, only space dependent) around the conductor in which charge flows. If the charged particle accelerates, in addition to electric field it also produces magnetic field. Both electric and magnetic fields are time varying fields. Since the electromagnetic waves are transverse waves, the direction of propagation of electromagnetic waves is perpendicular to the plane containing electric and magnetic field vectors.

Any oscillatory motion is also an accelerating motion, so, when the charge oscillates (oscillating molecular dipole) about their mean position as shown in Figure 5.9, it produces electromagnetic waves.

Suppose the electromagnetic field in free space propagates along z direction, and if the electric field vector points along y axis then the magnetic field vector will be mutually perpendicular to both electric field and the propagation vector direction, which means

$$E_y = E_\circ \sin\left(kz - \omega t\right)$$

$$B_x = B_\circ \sin\left(kz - \omega t\right)$$

UNIT-5(XII-Physics_Vol-1).indd 291 04-03-2019 11:05:48